



## This wearable exosuit reduces muscle fatigue



Vanderbilt University engineers have determined that their back-assist exosuit, a clothing-like device that supports human movement and posture, can reduce fatigue by an average of 29 to 47 per cent in lower back muscles. The exosuit's functionality presents a promising new development for individuals who work in physically demanding fields and are at risk for back pain, including medical professionals and frontline workers.

The research, led by assistant professor of Mechanical Engineering Karl Zelik and recent PhD graduate and primary author Erik Lamers, used surface electromyography techniques to measure changes in low back muscle fatigue in male and female participants, who were given physical tasks to perform both with and without the exosuit.

The wearable technology developed by Zelik's team may conjure images of Iron Man's suit, but it does not rely on motors or batteries. Instead, the low-profile elastic exosuit applies assistive forces that cooperate with the low back extensor muscles to relieve strain on the muscles and spine and help reduce injury risks.

This study showed that wearing the exosuit made holding a 15kg weight (the average weight of a four-year-old child) less tiring on the back than holding an 11kg weight (the average weight of an eighteen-month-old baby) without the exosuit.

*Exosuit male fit (Credit: Vanderbilt University)*



## Google optimises Assistant for people with disabilities

Google has announced a partnership with Tobii Dynavox, a US-based accessibility firm. Under the partnership, Google will integrate its Assistant into the company's apps and hardware. The objective of the partnership is to allow people with disabilities such as autism, ALS, and cerebral palsy to use Google Assistant features.

With the Assistant integration, users with disabilities can now use Tobii's hardware and applications to control smart home devices such as TVs and lights that are compatible with

the Google Home app. Users can also use the service to get Google Assistant to answer questions such as "What is the weather today?" and "What's on the calendar for today?," reports CNET.

The latest partnership comes months after Google launched new accessibility features for Android users. The new set of features also includes Assistant-based Action Blocks. Google also updated Sound Amplifier, Google Maps, and Live Transcribe features for users with disabilities.



## Vehicle damage insurance using artificial intelligence

Admiral Seguros, the Spain-based subsidiary of global insurer Admiral, has announced that it is using artificial intelligence (AI) to create damage valuations and generate an offer of immediate payment on damaged vehicles.

By using an AI solution developed by the technology

company Tractable, Admiral Seguros aims to accurately evaluate vehicle damage with photos sent through a Web application. The app, via the AI, completes the complex manual tasks that an advisor would normally perform and produces a damage assessment in seconds, often without